

Hans-Peter Marshall

Cryosphere Geophysics And Remote Sensing (CryoGARS) group
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Education

- 2005 Ph.D, Civil Engineering (Geotechnical), GPA: 3.9/4.0
Institute of Arctic and Alpine Research, University of Colorado at Boulder
Thesis: *Snowpack spatial variability: towards understanding its effect on remote sensing measurements and snow slope stability*
- 1999 B.S., Physics with honors, Geophysics Minor (1st in UW history), GPA: 3.6/4.0
University of Washington
Thesis: *Wet snow densification*
- 1994 Nathan Hale High School, Seattle, Washington, GPA: 3.9/4.0

Selected Professional Experience

- 08/2018-8/2021 Project Scientist, NASA SnowEx Mission
- 08/2016-07/2018 Co-lead of Field Campaign, NASA SnowEx 2017 Mission
- 08/2022-present Professor, Department of Geosciences, Boise State University
- 08/2013-08/2022 Assoc. Professor, Department of Geosciences, Boise State University
- 08/2008-08/2013 Assist. Professor, Department of Geosciences, Boise State University
- 07/2007-05/2020 Geophysical Engineer (Expert Consultant), U.S. Army Corps of Engineers
Cold Regions Research and Engineering Lab (CRREL), Hanover, New Hampshire

Research Interests

Snow science and glaciology, using geophysics, engineering, and remote sensing.
Spatial variability of snow, snow and ice mechanics, snow slope stability, meltwater dynamics
Computational statistics, geostatistics, compressed sensing, instrumentation development

Selected Honors and Awards

- 2020 BSU Faculty Excellence Award
- 2019 NASA Instrument Incubator Program Award, NASA, 1st in Idaho history
- 2019 2018 WRR Editors' Choice Award: Hedrick et al., 2018
- 2018 NASA Robert H. Goddard Team Award: SnowEx Year 1 Organizing Team
- 2012 Timeless Award, University of Washington, 1 of 150 College of Arts and Sciences alumni
awarded from UW history on 150th yr anniv. (1861-2011), 1 of 3 from Physics Department
- 2010-13 New Investigator Program Award, NASA, 1st in Idaho history
- 2010 Young Scientist Award, Cryosphere Section, American Geophysical Union
- 2009 Top Reviewer Award, Cold Regions Science and Technology (CRST)
- 2008 U.S. Young Scientist Award, International Union of Radio Science (URSI)

Selected Recent Peer-Reviewed Publications

[h-index=32, i10-index=68, citations=3175]

- 2023 Hoppinen, Z., Johnson, J. B., Capelli, A., Anderson, J. H., and **Marshall, H.P.** (2023). Snow attenuation of infrasound signals and wind noise. *Seismological Research Letters*, doi:10.1785/0220230048
- Rudisill, W., Flores, A., Marshall, H.-P., Siirila-Woodburn, E., Feldman, D., Rhoades, A., Xu, Z., and Morales, A. (accepted). Cold-season precipitation sensitivity to microphysical parameterizations: Hydrologic evaluations leveraging snow lidar datasets. *Journal of Hydrometeorology*
- Aberle, R., Enderlin, E. M., Marshall, H.-P., Kopera, M., and Meehan, T. G. (2023). Assessing controls on ice dynamics at crane glacier, antarctic peninsula, using a numerical ice flow model. *Journal of Glaciology*, 69(277):1109–1124
- Bonnell, R., McGrath, D., Hedrick, A. R., Trujillo, E., Meehan, T. G., Williams, K., Marshall, H.-P., Sexstone, G., Fulton, J., Ronayne, M. J., Fassnacht, S. R., Webb, R. W., and Hale, K. E. (2023). Snowpack relative permittivity and density derived from near-coincident lidar and ground-penetrating radar. *Hydrological Processes*, 37(10):e14996
- McDowell, I. E., Keegan, K. M., Wever, N., Osterberg, E. C., Hawley, R. L., and **Marshall, H.P.** (2023). Firn core evidence of two-way feedback mechanisms between meltwater infiltration and firn microstructure from the western percolation zone of the greenland ice sheet. *Journal of Geophysical Research: Earth Surface*, 128(2):e2022JF006752
- Hojatimalekshah, A., Gongora, J., Enterkine, J., Glenn, N. F., Caughlin, T., **Marshall, H.P.**, and Hiemstra, C. A. (2023). Lidar and deep learning reveal forest structural controls on snowpack. *Frontiers in Ecology and the Environment*, 21(1):49–54
- Tarricone, J., Webb, R., **Marshall, H.P.**, Nolin, A., and Meyer, F. (2023). Estimating snow accumulation and ablation with l-band insar. *The Cryosphere*, 17(5):1997–2019
- 2022 Lievens, H., Brangers, I., **Marshall, H.P.**, Jonas, T., Olefs, M., and DeLannoy, G. (2022). Sentinel-1 snow depth retrieval at sub-kilometer resolution over the European Alps. *The Cryosphere*, 16(1):159–177, doi.org/10.5194/tc-16-159-2022
- Lund, J., Forster, R., Deeb, E., Liston, G., Skiles, M., and **Marshall, H.P.** (2022). Interpreting sentinel-1 sar backscatter signals of snowpack surface melt/freeze, warming, and ripening, through field measurements and physically-based snowmodel. *Remote Sensing*, 14:4002, <https://doi.org/10.3390/rs14164002>
- McGrath, D., Bonnell, R., Zeller, L., Olsen-Mikitowicz, A., Bump, E., Webb, R., and **Marshall, H.P.** (2022). A time-series of snow density and snow water equivalent observations derived from the integration of gpr and uav sfm observations. *Frontiers in Remote Sensing*, 3:886747, doi:10.3389/frsen.2022.886747